TECHNICAL REPORT

Technical Report: Data Minimisation for student-Facing and Research Services

Organisation: RedKross Research Foundation
Domain: Privacy & Data Protection
Project Title: Data Minimisation: A Practical Privacy Guide for Students and Research Services
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1. Abstract

This project focuses on data minimisation in student-facing services and research activities. The objective is to explain how unnecessary personal information can be identified and reduced while ensuring that services and research continue to meet their intended purposes.

The project deliverables include a practical guide, a data minimisation checklist and three fictional before-and-after examples. The report documents the research approach, the principles used to develop these materials, their intended use and the limitations of the work.

2. Project Objective

The main objective is to develop practical privacy-awareness material that helps students and research teams collect only the personal information they genuinely need.

The specific objectives are:

- Explain the purpose of data minimisation.
- Identify unnecessary data fields in common student and research activities.
- Provide guidance on privacy notices, consent, retention, access control and safe deletion.
- Develop a checklist for reviewing data collection practices.

- Demonstrate data reduction through three fictional scenarios.
- Encourage responsible handling of personal information throughout its lifecycle.

3. Background Research

Data minimisation is an important privacy principle. It encourages organisations to limit personal data to what is adequate, relevant and necessary for the purpose of processing.

Collecting excessive information can increase the consequences of unauthorised access, data loss or misuse. It can also make records harder to maintain and increase storage and security responsibilities.

The project draws on the following sources:

1. **General Data Protection Regulation (GDPR), Article 5:** Provides the principle of data minimisation and related requirements concerning purpose limitation and storage limitation.
2. **Information Commissioner's Office (ICO):** Provides practical guidance on collecting only the information required for a stated purpose.
3. **NIST Privacy Framework:** Provides a structured approach to identifying and managing privacy risks.
4. **Digital Personal Data Protection Act, 2023 (India):** Provides an Indian legal reference relevant to the handling of digital personal data, subject to applicable commencement provisions and rules.

These sources provide the conceptual and practical foundation for the project. The guide is intended for awareness and does not constitute a formal legal compliance assessment.

4. Methodology

The project uses a document-based research and scenario-analysis approach.

Step 1: Identify the privacy problem

Common student-facing and research activities were considered, including workshop registration, research surveys and student support appointments.

Step 2: Review privacy principles

Relevant guidance was reviewed to identify practices involving purpose, necessity, transparency, access restrictions, retention and deletion.

Step 3: Develop the guide

The findings were organised into clear sections covering why information is collected, how to identify necessary fields, how information should be protected and when it should be deleted.

Step 4: Prepare the checklist

The checklist translates the guide's recommendations into questions that a team can use when reviewing a data collection process.

Step 5: Create before-and-after scenarios

Three fictional situations were developed to compare excessive collection with a more limited approach. Each revised scenario retains the information needed for its stated purpose while removing unrelated fields.

Step 6: Review the deliverables

The materials can be reviewed for clarity, consistency, relevance and completeness against the assigned task requirements. Formal user testing or independent legal review should be recorded only if actually performed.

5. Tools and Technologies

The project is primarily a research and documentation task.

The planned tools and resources are:

- Web-based research for consulting privacy guidance and official publications.
- A document editor for preparing the guide, checklist and technical report.
- Markdown for documenting the project in GitHub.
- Git and GitHub for version control, repository organisation and submission.

- A video-recording tool for the explanatory presentation.

No software application or automated data-minimisation system is claimed to have been developed as part of this documentation-focused project.

6. Work Performed and Deliverables

The project materials are structured as follows:

Deliverable 1 — Practical Guide: Explains the purpose of data collection, necessity, retention, access control, privacy notices, consent and secure disposal.

Deliverable 2 — Checklist: Provides review points for collection, transparency, storage, permissions, retention and deletion.

Deliverable 3 — Before-and-After Examples: Demonstrates data reduction in workshop registration, library-use research and student support.

Deliverable 4 — Technical Report: Documents the project's objective, research approach, methodology, expected use and limitations.

The materials are intended to help a student or research team review whether each requested data field has a justified purpose.

7. Findings and Evaluation

The scenario analysis illustrates several practical findings:

- Some forms request information that is unrelated to their stated purpose.
- A single suitable contact method may be sufficient for routine event communication.
- Anonymous responses or participant codes can reduce the need to retain direct identifiers in research datasets.
- Restricting access reduces unnecessary exposure of personal information.
- Defining retention periods helps prevent indefinite storage.

- A deletion process should consider copies, shared folders and backups, not only the original file.

The three scenarios provide qualitative demonstrations of how data collection can be reduced. They are illustrative examples, not measured results from an actual institutional audit.

8. Testing and Evaluation Plan

The following checks can be performed before final submission:

1. Confirm that the guide covers all required topics.
2. Verify that the checklist includes collection, notice, consent where appropriate, access, retention and deletion.
3. Confirm that exactly three before-and-after examples are included.
4. Check that each revised example retains the information necessary for its purpose.
5. Verify that the references lead to relevant sources.
6. Open the files from the GitHub repository to confirm they are readable.
7. Test the repository and video links using a separate browser session or account where appropriate.

These checks are a proposed evaluation plan. Mark each check as completed only after performing it, and document any issues discovered.

9. Limitations

- The project is based on general privacy guidance and fictional scenarios.
- No real student or research-participant records were used in the examples.
- The scenarios do not represent an audit of an actual institution.
- The project does not establish that an organisation complies with every applicable legal requirement.
- Retention periods, legal bases and consent requirements depend on the particular purpose, applicable law and institutional policy.

- No measured reduction in collected data or improvement in privacy risk is claimed without real testing and supporting evidence.

10. Recommendations

1. Require a documented purpose for every personal-data collection activity.
2. Review forms regularly and remove unnecessary fields.
3. Provide clear privacy notices before or at the point of collection.
4. Use anonymous or pseudonymous research data where appropriate.
5. Apply least-privilege access controls.
6. Establish documented retention and secure-deletion procedures.
7. Train students, researchers and staff in responsible data handling.
8. Review new collection activities for privacy risks before launch.
9. Escalate sensitive or legally uncertain cases to the responsible privacy, legal or research-ethics contact.

11. Conclusion

This project provides practical materials for understanding and applying data minimisation in student-facing and research services. The guide explains key privacy practices, the checklist supports routine reviews, and the fictional examples demonstrate how unnecessary data fields can be removed.

The central conclusion is that responsible data collection requires a clear purpose, a justified selection of fields, appropriate safeguards and a defined end to retention. Applying these practices can reduce avoidable privacy exposure while preserving the information needed to provide services and conduct valid research.

12. References

1. European Union. *Regulation (EU) 2016/679, General Data Protection Regulation, Article 5.* https://eur-lex.europa.eu/eli/reg/2016/679/oj

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3. National Institute of Standards and Technology. *NIST Privacy Framework.* https://www.nist.gov/privacy-framework
4. Ministry of Electronics and Information Technology, Government of India. *Digital Personal Data Protection Act, 2023.* https://www.meity.gov.in/data-protection-framework